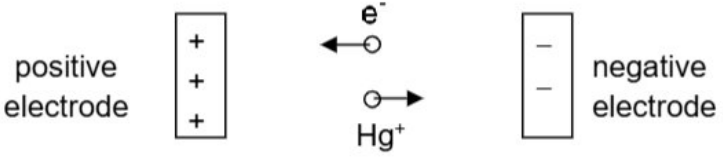
20 A high potential difference is applied between the electrodes of a mercury (Hg) discharge tube such that the gas between the electrodes is ionized.



In each minute, 1.5×10^{15} electrons and 4.4×10^{15} positive mercury ions (singly charged) move past a cross-section of the tube in opposite directions.

What is the total current between the electrodes?

A 8.0 μA **B** 16 μA **C** 0.46 mA **D** 0.94 mA